Nixor O Levels
Test Week 2
Biology
Grade 10

41

Student Name: _____

Student ID: _____

Section : _____

Time: 60 minutes

Max marks: 44

Please read and follow the instructions below:

- Please write your name, section and ID on the question paper.
- Do not turn over the paper unless instructed to do so.
- Only write in blue or black ink.
- Use of correction fluid is strictly prohibited.
- If you have any questions, raise your hand to ask the invigilator.
- Omission of working will result in loss of marks.
- Use a pencil for making straight lines.
- Answer all questions.
- Calculators are allowed.

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1 (a) Write a balanced chemical equation for photosynthesis.

$6CO_2 + 6H_2O \xrightarrow[\text{chlorophyll}]{\text{sunlight}} C_6H_{12}O_6 + 6O_2$ [3]

3

A student investigated the effect of light intensity on the rate of photosynthesis of algae.

Fig. 2.1 shows the apparatus set up for the investigation.

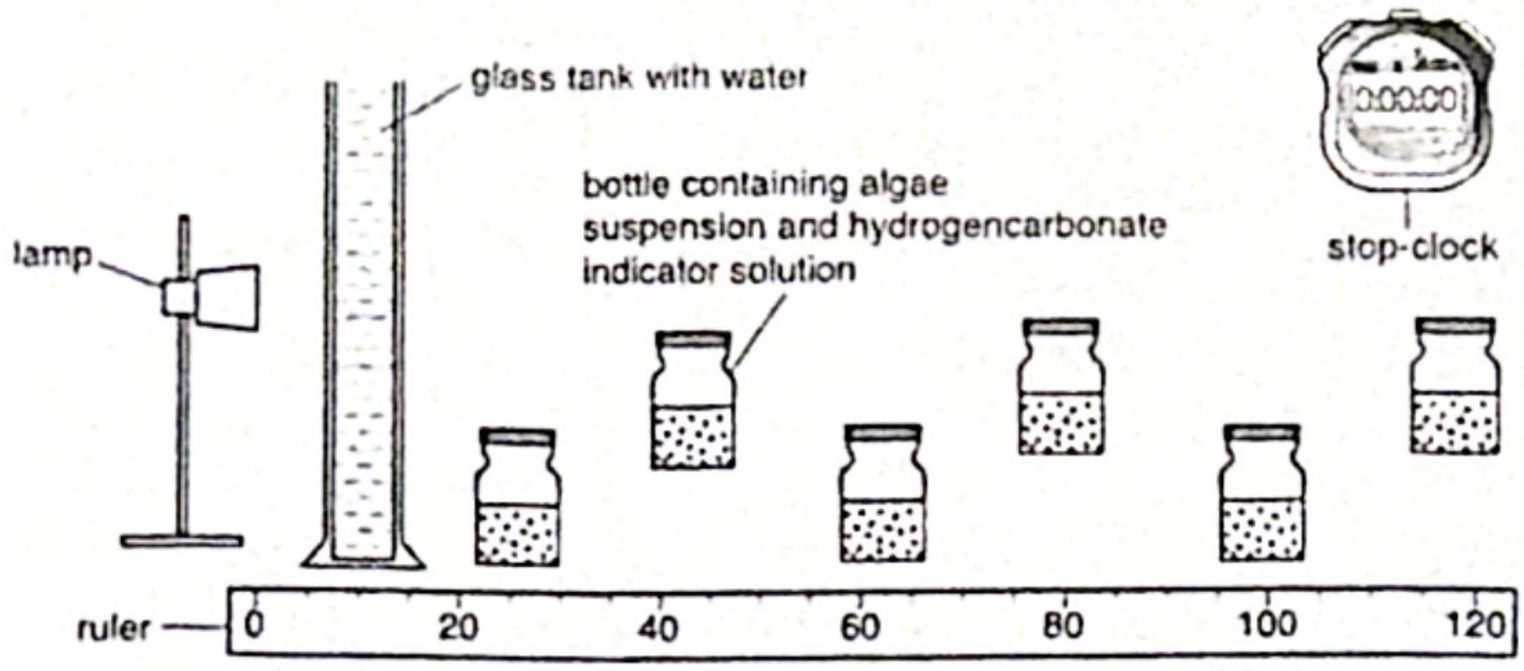


Fig. 2.1

(b) Suggest why a glass tank with water was placed between the lamp and the bottle in the investigation.

To spread out the lights from the lamp and to make it so that from the light does not cause changes in temperature. so it hits all plants [1]

1

(c) The hydrogencarbonate indicator solution changes colour when the pH changes. At pH 8.4 it is red, at pH 7.6 it is yellow and at pH 9 it is purple.

Predict the colour of the hydrogencarbonate indicator solution in the bottle nearest the lamp at the end of the investigation. Explain your answer.

colour prediction purple

explanation Since it's nearest to the lamp it gets most amount of light and hence rate of photosynthesis increases which takes in carbon dioxide and gives out oxygen making the solution more alkaline and hence turning it purple.

3

[3]

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(d) The student's results are shown in Fig. 2.2.

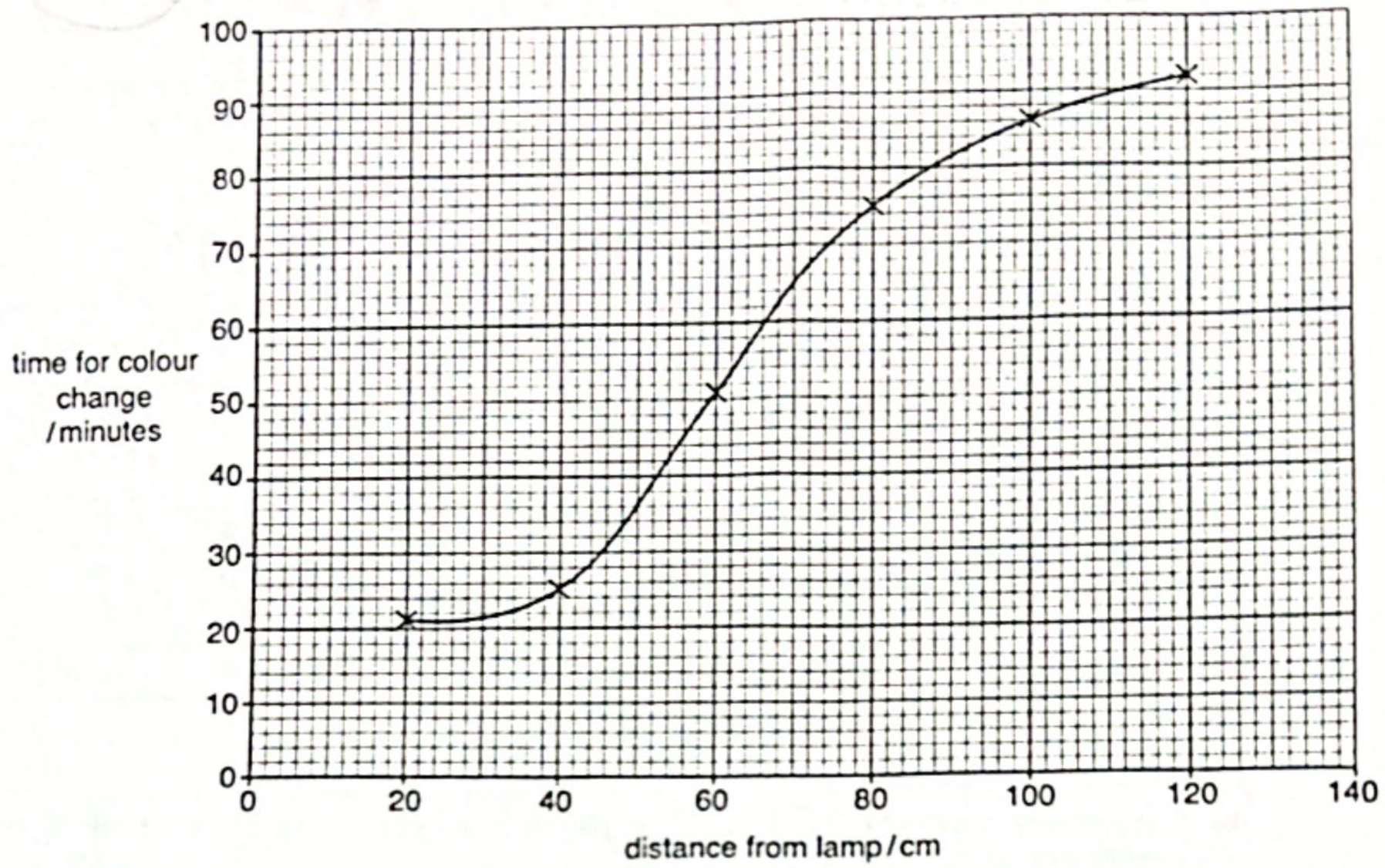


Fig. 2.2

Describe and explain how the rate of photosynthesis is affected by light intensity.

The more light intensity there is the more light is absorbed in the chloroplasts of plants. As light is needed for Photosynthesis as it is converted to chemical energy. So the more light intensity there is, the more chemical energy for photosynthesis is available and hence the rate of photosynthesis increases. Similarly if light energy were to decrease then less energy for Photosynthesis at first and so photosynthesis is reduced. This shows direct proportionality. At distance from lamp = 20cm the light intensity was the most and so it took the least time which was 21 minutes to change colour and hence photosynthesis was happening at a faster rate compared to 120 cm from lamp where it took 92 minutes to change colour. (4) [5]

[Total: 12]

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2.

Photosynthesis is the process by which plants obtain nutrition.

(a) Write a balanced chemical symbol equation for photosynthesis.

2

(2)

$6CO_2 + 6H_2O \xrightarrow[\text{Chlorophyll}]{\text{sunlight}} C_6H_{12}O_6 + 6O_2$

(b) Leaves can be tested for starch to show that photosynthesis has taken place.

This is the method given in a biology textbook.

1. Remove the leaf you want to test from the plant.
2. Half fill a 250 cm^3 beaker with water and boil the water using a Bunsen burner.
3. Using a pair of forceps, hold the leaf in the boiling water for 20 seconds.
4. Turn the Bunsen burner off.
5. Using the forceps, push the leaf to the bottom of a boiling tube and cover it with ethanol.
6. Place the boiling tube in the beaker of very hot water. The ethanol will boil.
7. When the leaf is colourless, remove it from the boiling tube and wash it in cold water for a few seconds.
8. Place the leaf flat on a white tile.
9. Add dilute iodine solution with a pipette, making sure the whole leaf is covered.
10. Any starch present will react with the iodine solution.

(i) Explain a safety precaution that should be taken when carrying out this test.

(2)

Turn the bunsen burner off before carrying the boiling tube with ethanol as ethanol is a flammable substance and can easily catch on fire creating safety hazards. So Extinguishing the flame reduces risk. 2

(ii) State the purpose of step 3.

(1)

It is to kill the cytoplasm, denature ~~denatures~~ the enzymes and makes the leaf more permeable to iodine so that iodine test for starch can be performed on it. 1

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(iii) State the purpose of step 6.

(1)

It is to boil at the ethanol which will ~~take~~ dissolve out the chlorophyll too allowing us to see the colour changes later on in the iodine test.

(c) Experiments can be done to demonstrate that light, chlorophyll and carbon dioxide are needed for photosynthesis.

Testing leaves for starch is the final step in these experiments.

Explain what other steps need to be taken to demonstrate that each of these factors is needed for photosynthesis.

(i) light

(2)

You need to destarch ~~you need to destarch~~ on the plant and then put a piece of aluminium foil over some part of the plant. Then you need to ~~let it~~ put it in sunlight for 5-6 hours. Then perform the steps to allow iodine to come in and chloroplasts (chlorophyll) to be shown.

(ii) chlorophyll

(2)

You need to destarch a variagated leaf and then put it in sunlight for 5-6 hours to photosynthesis. Then put it in boiling water to make iodine permeable and use boiling ethanol to remove chloroplyll.

(iii) carbon dioxide

(2)

Firstly you need to destarch the plants, then put a transparent plastic bag with ~~sodium~~ restricts atmospheric gasses and put soda lime in one of the tests to absorb the carbon dioxide and Sodium hydrogen carbonate in the other to Produce carbon dioxide. Then after a few hours ~~take~~ a leaf from each and ~~dip it in~~ dipping in boiling water and submerge in boiling Ethanol to ~~brown~~ chlorophyll and make it ~~permeable~~ permeable to iodine.

(Total for Question = 12 marks)

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3. Fig. 1.1 **A** shows a cell from the lining of the alimentary canal.

Fig. 1.1 **B** shows a cell from the lining of a kidney tubule.

Both cells absorb substances into the blood.

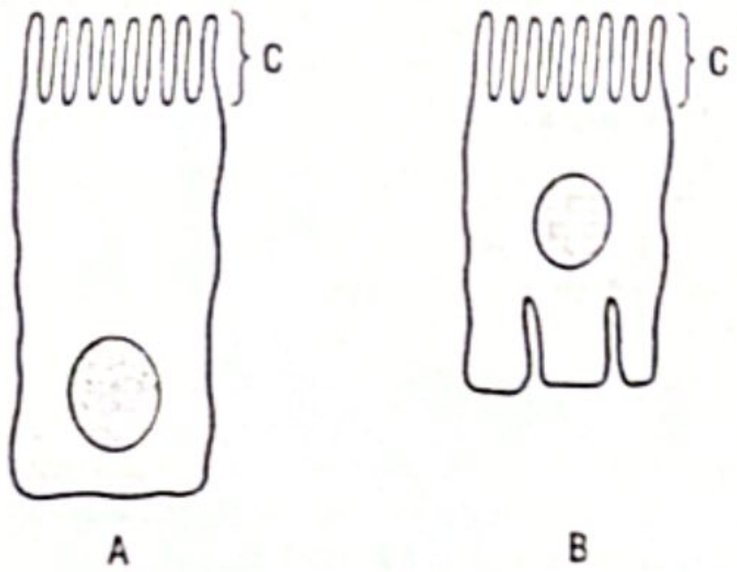


Fig. 1.1

(a) Name the structures labelled **C** on the cells in Fig. 1.1.

~~epithelial cells~~ villi ..... [1]

(b) List three substances that are absorbed by **both** cells shown in Fig. 1.1.

1 glucose .....

2 amino acids .....

3 fatty acids and glycerol ..... [3]

(c) Explain how **both** cells shown in Fig. 1.1 are adapted for absorption of substances into the blood.

they have villi which are tiny finger like projections which increase the surface area and absorbance absorption rate. they also .....

..... [2]

(d) Name the part of the alimentary canal that is lined by the cells shown in Fig. 1.1 **A**.

small intestine ..... [1]

[Total: 7]

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4.

A student investigated the nutrient content of three types of drink:

- drink **A**
- drink **B**
- drink **C**.

The student used these methods for the three tests on each of the drinks:

Test 1 Testing for reducing sugars:

- Label three test-tubes **A**, **B** and **C**.
- Put 1 cm^3 of drink **A** into test-tube **A**.
- Put 1 cm^3 of drink **B** into test-tube **B**.
- Put 1 cm^3 of drink **C** into test-tube **C**.
- Add 1 cm^3 of Benedict's reagent to each test-tube.
- Put all three test-tubes into a hot water-bath.
- Start the stop-clock and leave the test-tubes in the water-bath for five minutes.
- After five minutes, remove the test-tubes from the water-bath.
- Observe and record the colours in each test-tube.

Test 2 Testing for starch:

- Place two drops of each drink onto a white tile.
- Add two drops of iodine solution to each drink sample.
- Observe and record the colours on the tile for each sample.

Test 3 Testing for protein:

- Label three test-tubes **A**, **B** and **C**.
- Put 1 cm^3 of drink **A** into test-tube **A**.
- Put 1 cm^3 of drink **B** into test-tube **B**.
- Put 1 cm^3 of drink **C** into test-tube **C**.
- Add 1 cm^3 of biuret reagent to each drink sample.
- Observe and record the colours in each test-tube.

(a) The student's observations are shown in Fig. 1.1.

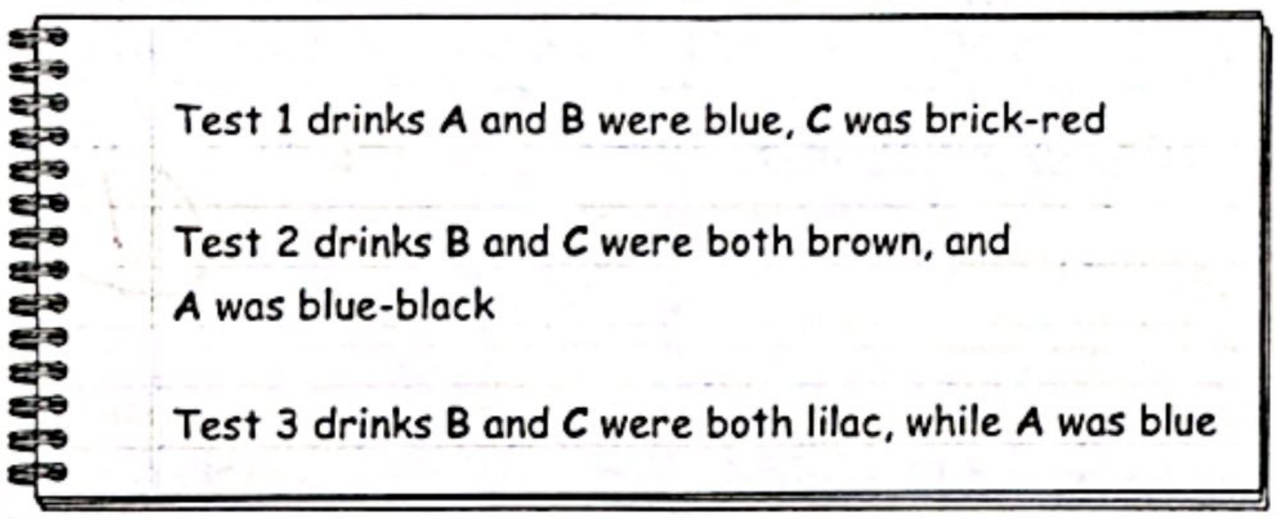


Fig. 1.1

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(i) Prepare a table to record the **colours observed** by the student for all three tests for each drink.

Do **not** include conclusions in your table.

	Test 1 (reducing sugar)	Test 2 (starch)	Test 3 (protein)
Drink A	blue	blue-black	blue
Drink B	blue	brown	lilac
Drink C	brick-red	brown	lilac

4

[4]

(ii) State which nutrients are present in each drink.

drink **A** starch

drink **B** protein

drink **C** reducing sugars and protein.

3

[3]

(iii) Identify **one** safety hazard associated with **Test 1**.

~~Water~~ Hot water could spill and cause skin ~~damage~~. Hot water used in the benedict's reagent could spill and cause burns or damage to skin. [1]

1

(b) The vitamin C content and the fat content of three other drinks **D**, **E** and **F** were determined.

It was found that:

- drink **D** contained vitamin C
- drink **E** contained fat
- drink **F** contained vitamin C and fat.

(i) State the reagent used when testing for vitamin C.

DCPiP solution. [1]

(ii) Describe the method for the emulsion test for fats.

First add the fat sample into a test tube and add ethanol to it. Shake it with well and then pour the solution in ~~a bottle~~ a test tube which has distilled water and shakes gently. If tiny drops of emulsion form then it has fats. [2]

(iii) The results for **one** of the drinks are shown in Table 1.1.

Table 1.1

test	observation
vitamin C	the solution is colourless
fat	a white emulsion has formed

Identify the drink from the results provided in Table 1.1.

drink F [1]

(iv) Explain how you identified the drink from the results provided in Table 1.1.

Vitamin C's DCPiP solution turns from ~~colourless~~ blue to colourless and fats emulsion test form a white emulsion when vitamin C and fat are present respectively. [1]
Drink F matches with the results.

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